Anne Marthe Sophie Ngo bibinbe https://ngobibibnbe.github.io/
Quebec, Canada anne.ngobibinbe@gmail.com
418 906 2807

EXPERIENCES

Research assistant 2022-Present
Laval University, Québec

- **Project1: Long-term identity aware multi-animal tracking** 2022-2024

  Proposal of a long-term tracking Framework based on HMM merging stochastic identity information with classic tracking by detection to solve the long-term tracking problem with a use case on livestock tracking. The model improved ByteTrack identity F1-score of 9% in our dataset and we see significant improvements proportional to the number of uncertain identifications (Oral at ***CVPR 2024*** *animal workshop accepted to IJCV).*

  Development of the PromptTrack library (https://pypi.org/project/PromptTrack/) for tracking any object based on prompt.

- **Project2: Animal behavior analysis** 2024

  Proposed an unsupervised pig aggression detection and entity localization in video by leveraging a Conv3D based autoencoder and tracking, achieving 60% of accuracy in a pig environment setting covering 3days of footage.

PhD Student (Visitor) 2025
University of Central Florida, Florida

(Ongoing) Proposing a zero-shot STAD model. The purpose is given a list of action prompts someone is interested in and a video; the model should provide tubelets concerned by those actions as well as the classes among the prompt Classes. The problem from a modeling perspective is an open vocabulary, multi-label classification with a tube head for tube proposal. Here I changed the architecture of Internvideo1 to add an open vocabulary head for embedding prediction and a tube proposal head for tube prediction. The loss is a mixture between cross input consistency, GIOU, L1 and presence Losses. The next step will be to add RL with GRPO to improve model performance based on some reward functions we are working on.

AI consultant Winter 2024-Present
Machinex, Québec

- Developed several models for factory sorting robots with a 95% precision and 90% recall
- Developed an evaluation pipeline for the sorting models with sanity check to identify problematic patterns.
- Developed test time adaptation models to improve performances of models on new client factories based on other data.
- Developed a model to estimate weight of materials based on 2D images with 6% MAPE 2% lot average error. This was a 13% MAPE improvement on former method based on weighted pixel average.

AI researcher, Anomaly detection on streaming data 2021-2022
CNRS/Pfeiffer-Vacuum, Clermont-Ferrand, France

- State of the art and Benchmark of anomaly detection methods in data streams and time series. (Accepted to IJCNN)
- Design and development of an explainable anomaly detection model for a real-time air contamination monitor.
- Proposed a method for detecting abnormal subsequences in data streams (Drag-stream, accepted at the ICDM 2022 conference).

Junior Machine learning engineer Winter 2021
Omdena, Californie, USA

- Development of a recommendation system to advise users on activities to reduce their biological age.

STUDY

PHD in computer vision applied to animal behavior tracking since 2022
Laval University, Quebec / UCF, Florida

SKILLS

Domain: Computer Vision, RL, VLM, LLM
Frameworks: Tensorflow, Pytorch
Technologies: Git, Docker, GCP, Vertex AI

SCIENTIFIC PUBLICATION

IJCV 2025 : oral CV4animals (CVPR 2024) 2024
An HMM-based framework for identity-aware long-term MOT from sparse and uncertain identification: use case on long-term tracking in livestock.
Anne Marthe Sophie Ngo Bibinbe, Patrick Gagnon, Jamie Ahloy-Dallaire, Eric R. Paquet

ICDMW 2022 2022
DragStream: An Anomaly and Concept Drift Detector In Univariate Data Streams,
Sophie Ngo Bibinbe, Abdoul Mahamadou, Michael Mbouopda, Engelbert Mephu Nguifo

IJCNN IEEE 2022 2022
A survey on unsupervised learning algorithms for detecting abnormal points in streaming data,
Sophie NGO BIBINBE, Michael MBOUOPDA, Raissa MBIADEU SALEU, Engelbert MEPHU

Short article: EGC 2022 2021
Benchmarking unsupervised methods for abnormal point detection in data stream,
Sophie NGO BIBINBE, Michael MBOUOPDA, Raissa MBIADEU SALEU, Engelbert MEPHU